

TdhGISnet API

Overview
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Introduction

The TdhGISnet API provides general network path analysis functionality in an easy to library with both C++ and Python interfaces. The functionality provided includes:

- The shortest path between any 2 points in the network.
- The shortest path between a starting point and any one of a set of potential end points (nearest location).
- The shortest paths from a starting point to all other points.
- The shortest path between a starting point and an entire set of intermediate points in a specified order.
- The shortest path between a starting point and an entire set of intermediate points in any order (traveling salesman).
- The shortest path between a set of specified points, where any of the points can be designated as a potential start or end point.

These operations may be particularly useful for a network where the path values (e.g. travel times) are frequently changing or where topology (connection data) is frequently changing.

These features are highlighted in compact demonstration source code that will get you going in the time it takes to download other packages.

Libraries and Header Files

The header files contain documentation for the included classes and functions. They can be found in the `cpp/include` directory of the package.

The executable library files are contained in subdirectories of the `bin` directory, associated with one of 3 compilers:

- `gcc` for linux
- `msvc` for mswin
- `mingw` for mswin

2 libraries are needed to perform path analysis:

- TdhCommon provides general data structure management, defined in the file `TRecordsNav0.h`, and geometry specific data structures, defined in the file `CadPrimitives0.h`.
- TdhPath performs the heaving lifting for path analysis with the interface defined in `TdhPath_Intf.h`

These libraries and header files are the same used for the TdhGISnet app, so they can support everything from a simple to a robust path analysis project.

The TdhGISnet_API library is used to perform database I/O requiring only standard sqlite. The interface is provided in the TdhGISnet_API.h header file. The database structure is the same used for the TdhGISnet app, so the app can be used to create input data and inspect results.

OpenStreetMap data can be imported using the TdhGISnet_API library, as documented in the TdhGISnet_API.h header file and the TdhGISnet help file. One caveat is that the TdhGISnet_API library cannot perform coordinate conversion, so the imported data will be in longitude/latitude coordinates. If coordinate conversion is needed, the TdhOGR app or the TdhOGR_API library, included with the TdhGISnet_API package, can be used.

Demonstration App

A demonstration app, contained in the source directories (cpp and python) of the distribution package, is intended show how to use the TdhGISnet API quickly and easily. It is intended to be run as a console app.

The demo needs access to the files in the TdhGISnet_API_demo_Data directory, included in the package. Both the cpp and python source code include a variable for the data directory, which should be set consistent with the placement of the directory by the user. The python source code includes a file, TdhGISnet_API_universal.py, which sets the directory for the library files.

Python Bindings

TdhGISnet_API now provides python bindings so that virtually all the functionality provided by the API is accessible through python.

Python must have access to the same executable libraries provided for C++, plus the libraries that provide the python bindings. All such libraries are provided in the bin directory of the package. Otherwise, nothing beyond standard python is required.

A series of python classes corresponding to the C++ API classes are provided in the python directory of the package. Documentation is provided with each function definition.

A python version of the demonstration code, TdhGISnet_API_demo.py, is provided in the python directory.

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